INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

VEST, COOLING (WITH FREEZABLE GEL STRIPS)

The General Services Administration has authorized the use of this Commercial Item Description.

Abstract. This document covers vest, cooling (with freezable gel strips). The vests are intended to be worn in heat stress conditions to extend Physiological Heat Exposure Limits (PHEL"s) of the Navy personnel.

Salient Characteristics. The vest shall be made of flame retardant treated cotton, reverse twill. Six pockets (three each front and back, full width of the vest) are used to hold the freezable gel strips. The pockets are insulated with a polyolefin/polyester batting, enclosed by two layers of metallized polyolefin, and lined with nylon tricot. The pockets are fastened closed with hook and loop fastener tape. The sides of the vest are open, with four hook and loop fastener tape adjustment straps. The straps and the hook tapes on the straps are secured by cross-box stitching. All outer edges of the front

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FSSD, 2800 South 20th Street, Philadelphia, PA 19101-8419, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8415

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and back are bound with bias cut basic material. All seams shall be double needle stitched, except binding. Each vest requires six freezable gel strips which have been doubled bagged in polyethylene.

Materials.

The material for the front, back, outer layer of pockets, adjustment straps and binding shall be preshrunk 100% cotton reverse twill cloth with a minimum thread count of 105 warp and 50 filling yarns. The minimum weight of the cloth shall be 8.5 (± 0.5) ounces per square yard and shall be able to meet the flame resistance requirements of not more than 2.5 seconds average time of after flame and a char length of not more than 5.0 inches in both the warp and filling directions when tested in accordance with TM 5903 of FED-STD-191. The color shall be Khaki. The fabric shall not shrink nor elongate more than 3 percent.

Lining material. The material for lining the inner layer of the pocket shall be nylon tricot, weighing approximately 5-1/2 ounces per square yard. The color shall be white, natural or bleached white.

Hook and loop fastener tape. The hook and loop fastener tapes for the adjustment straps and pocket openings shall conform to the following:

Hook tape - 6.5 mil (220 denier) or 8.0 mil (330 denier) with non-melting aramid ground.

Loop tape - non-melting aramid.

The color shall be natural, white, black or approximate the basic material. The hook and loop fastener tape for the pocket openings shall be 5/8 inch wide and for the adjustment straps 1-1/2 inches wide.

Thread. The thread for seaming and stitching the vest shall be a spun staple aramid thread, nominal 40-45 Tex. The color shall approximate the basic material.

Insulating material. The order of insulating material for the pocket assembly shall consist of the following:

1 - layer of basic material
2 - layer of metalized mylar assembly
3 - layer of 65% polyolefin/35% polyester batting
4 - layer of metalized mylar assembly
5 - layer of nylon tricot

NOTE: The non-metalized surface (when applicable) shall be against the batting layer.

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Metalized mylar assembly. The metalized mylar assembly shall consist of a layer of aluminum sheet fused or vacuum deposited to a sheet of mylar. The thickness of the aluminum sheet with mylar shall be 0.001 ± 0.0005 inch and the weight shall be 0.5 ± 0.05 ounces per square yard. The metalized mylar may be fused to a nonwoven for added strength or two layers of metalized mylar may be assembled together with netting between the layers and on the outsides and stay-stitched together. The thickness of the metalized mylar assembly shall be 0.006 ± 0.0005 inch for the nonwoven assembly and 0.009 ± 0.0005 inch for the netting assembly. The total weight for the nonwoven assembly shall be 1.8 ± 0.1 ounce per square yard and for the netting assembly 1.9 ± 0.1 ounce per square yard. Any nonwoven or netting is acceptable provided the overall thickness and weight requirements of the metalized mylar assembly are met.

Gel strips. Nine (± 0.5) ounces of plastic like gel (semi-solid starch base gel) shall be sealed into a 0.004 mil low temperature polyethylene bag. A preservative shall be added to the gel to inhibit bacterial growth. The plastic like gel shall be able to withstand temperatures below -20°C and shall not leak from the bags if punctured. The plastic like gel shall have a heat of fusion of 55.8 calories/gram when tested in accordance with ASTM E 793. The seams of the polyethylene bags shall be heat sealed and pack shall measure approximately 5 by 5 inches. Three gel bags shall be sealed into a strip with heat sealed seams along the fours sides and between each bag. When a tub bag is used, the top and bottom does not have to be sealed. The gel strips shall measure approximately 5-1/2 by 15 inches (±1/2 inch). The seams of the inner bag shall be sealed so that no air is trapped within the bag. The polyethylene material for both the inner and outer pack shall be a minimum of 0.004 mil (± 20 % tolerance) thickness.

Size. The vest shall be one size only and shall meet the dimensions shown in Figure 1. Adjustments shall be made hook fastener tape attached to the side adjustment strap matching with the loop fastener tape stitched to the top of the two lower front pockets.

Patterns and pattern parts. The Government shall furnish the patterns for the one size. The following is a list of patterns parts and number of cut parts of each:

Nomenclature	Cut parts
Front	1
Front pocket 1/	1
Back	1
Back pocket 1/	1
Strap	8

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1/ The pattern for the nylon lining and insulating material shall be the same at the front and back pockets.

Instruction label for vest. A commercial type label is acceptable provided that the following information is included. The label shall be caught in the neck binding at the inside back.

Donning Instructions

1. Insert frozen packs into each of the pockets. Close pocket.
2. Slip vest over head. Adjust side straps for a close, comfortable fit. One layer of clothing shall be worn under the vest.

Cleaning Instructions
Remove gel strips
Close hook and loop fastener tapes
Machine wash warm or cold water only
Tumble dry cool or hand dry

DO NOT USE HEAT IN WASHING AND DRYING OF THIS GARMENT
or
Dry clean using no heat

Contract number and NSN for vest. The vest shall have a label containing the contract number and NSN caught in the stitching of the commercial label or stitched in the neck edge binding at the inside back.

Instruction label for gel strips. A commercial type label shall be stamped on the gel strips provided the following information is included:

DATE OF MANUFACTURE: 07/92 (EXAMPLE)
CONTRACT NO.: DLA100-92-000-0000 (EXAMPLE)
NSN: 8415-000-0000 (EXAMPLE)

Two years from the date of manufacture the gel strips shall be weighed. If the weight is less than 8.5 ounces discard.
If the weight is equal to or greater than 8.5 ounces put back and reweigh every two years.

Workmanship. After assembly, the vest shall be thoroughly cleaned and all loose threads, lint and foreign matter removed. The vest shall be uniform in quality and shall be free from

irregularities or defects which adversely affect performance, reliability or durability.

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Product Demonstration Model. Offerers are required to submit vests or gel strips sets, as applicable, of their intend production and style. Product Demonstration Model will consist of units independently developed. Product Demonstration Models will be analyzed and evaluated by the government. Evaluation criteria and basis for award are contained in the solicitation.

Product Demonstration Model will be retained by the government. One (1) Product Demonstration Model approved by DPSC under the resultant contract will serve as a manufacturing standard for the corresponding production items delivered under this contract. Offerers are advised that the Product Demonstration Model does not relieve the successful offerer of this responsibility to perform in accordance with the Commercial Item Description specified above.

Responsibility for inspection. The supplier is responsible for assuring the product conforms to the requirements. Vest lots will be inspected to determine conformance with the Commercial Item Description and the successful offerer's technical proposal as negotiated. Acceptability of vest lots will be determined by the use of "Sampling Procedures and Tables for Inspection by Attributes", ANSI Z1.4, published by the American Society for Quality Control. Sampling procedures will be designated by the Government at the time of award. The sample unit is one vest.

Examination of the end item. The vest or gel strips as applicable, shall be examined for the defects listed below. The lot size shall be expressed in units of one completed vest or one set of gel strips (six strip per set), as applicable. The sample unit shall be one vest or one set of gel strips. The inspection level shall be II and the AQL shall be 1.5 for major defects and 6.5 for major and minor defects combined.

Defects	Classification Major	Minor
VEST		
I. MATERIAL DEFECTS AND WORKMANSHIP DAMAGE		
a. Component part full, tight or twisted	X	
b. Open or puckered seam	X	
c. Any size hole, cut, tear, smash or misweave	X	
d. Missing yarn or broken end		X
e. Loose yarn		X
f. Visible mend, needle chew or irregular stitching		X
g. Pleat or crease		X

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Defects	Classification Major	Minor
II. SHADE VARIATIONS		
a. Parts badly shaded from other parts or within parts		X
b. Color of thread, hook fastener tape or loop fastener tape not a good match with material when applicable		X
III. CLEANNESS		
a. Any spot or stain clearly noticeable		X
b. Thread ends not trimmed		X
IV. DESIGN AND ASSEMBLY		
Not cut or fabricated in accordance with the approved style/ product demonstration model	X	
V. SEAM AND STITCHING		
a. Seams puckered, twisted, pleated or any part of vest caught in an unrelated operation	X	
Any open seam, loose stitch tension or seam, stitch type and stitches per inch not in accordance with the approved style/ productdemonstration model	X	
VI. GENERAL		
Any defect not otherwise classified herein but at variance with the approved style/ productdemonstration model	X	
VII. POCKETS		
a. Not large enough for packs	X	
b. Opening does not stay closed	X	
c. Not uniform in size and location	X	
VIII. BINDING		
a. Binding omitted, twisted or not securely attached	X	
b. Binding not bias cut	X	
c. Binding not made of basic material	X	

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Defects	Classification Major	Minor
IX. SIDE STRAPS		
a. Incorrect width, length, number of piles or hook fastener tape not secured	X	
b. Strap not uniformly constructed or not constructed using double rows of stitching	X	
c. Not attached with cross boxstitch		X
X. LABEL		
a. Label missing, incorrect or illegible		X
XI. MEASUREMENTS		
Any part of garment not in accordance with tolerances on figure or garment measurement not in accordance with figure	X	
GEL STRIPS		
XII. MATERIAL DEFECTS AND WORKMANSHIP DAMAGES		
a. Low temperature polyethylene bag not 0.004 mil thick	X	
b. Welded seams open, unsealed, or may be easily pulled apart	X	
c. Evidence of moisture between inner and outer bag	X	X
d. Contents leak from bag when punctured	X	
e. Contents do not perform in accordance with requirements	X	
f. Bag material incorrect	X	
XIII. CLEANNESS		
a. Residual dust or solution		X
XIV. DESIGN AND ASSEMBLY		
Not fabricated in accordance with the approved style/product demonstration model	X	
XV. GENERAL		
Any defects not otherwise classified herein but at variance with the approved style/product demonstration model	X	

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XVII. MEASUREMENTS

Any part not in accordance with tolerances stated	X

Regulatory Requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

Packaging. Each vest shall have the fastener tape engaged, pockets closed and inside of front against inside of back. Fold each vest along top of middle pocket on front and back. Vest should measure approximately 15-1/2 by 21 inches.

PACKING

Vest. Six vests packaged as specified above shall be packed in a fiberboard shipping container, assembled, closed and reinforced according to type CF, class domestic, variety DW, grade 200, size 2A of MIL-B-17757. The vest shall be reversed from top to bottom within the box. The fiberboard for the liner shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Any voids in the box shall be filled with tissue paper.

Gel strips. Eighteen of the gel strips shall be packed in a fiberboard shipping container, assembled, closed, and reinforced according to type CF, class domestic, variety DW, grade 200, size 1A of MIL-B-17757. The fiberboard liner shall conform to type CF, class domestic, of MIL-B-17757. The gel packs shall be packed as follows: one in length, two in width and nine in depth. The container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper. Any voids in the box shall be filled with tissue paper.

Marking. In addition to any special marking required by the contract or order, unit packs, intermediate or exterior containers and palletized unit loads shall be marked in accordance with the requirements of MIL-STD-129.

Palletization. When specified in the contract, items packed as specified in above, shall be palletized on 4-way entry pallet in accordance with load type Ia of MIL-STD-147.

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NOTES

Approved styles. The following styles meet the requirements of this commercial item description:

Steele Inc.

Vest: Model No. SA 1140FR
Pack: Model No. SA 1149

Insulating material. The insulating material, 65% polyolefin/35% polyester is available as Thinsulate CS 150 from: 3-M Co. of St. Paul, MN 55144-1000.

Preservation, Packing, and Marking. The preservation, packing, and marking shall be as specified in the contract or order.

Source of Government Documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

Sources of Non-government Documents.

ANSI/ASQC Z1.4 - Sampling Procedure and Tables for Inspection by Attributes

(Applications for copies should be addressed to American National Standards Institute, 1430 Broad, New York, NY 10018-3308.)

ASTM E 793 - Heats of Fusion and Crystallization by Differential Scanning Calorimetry

(Applications for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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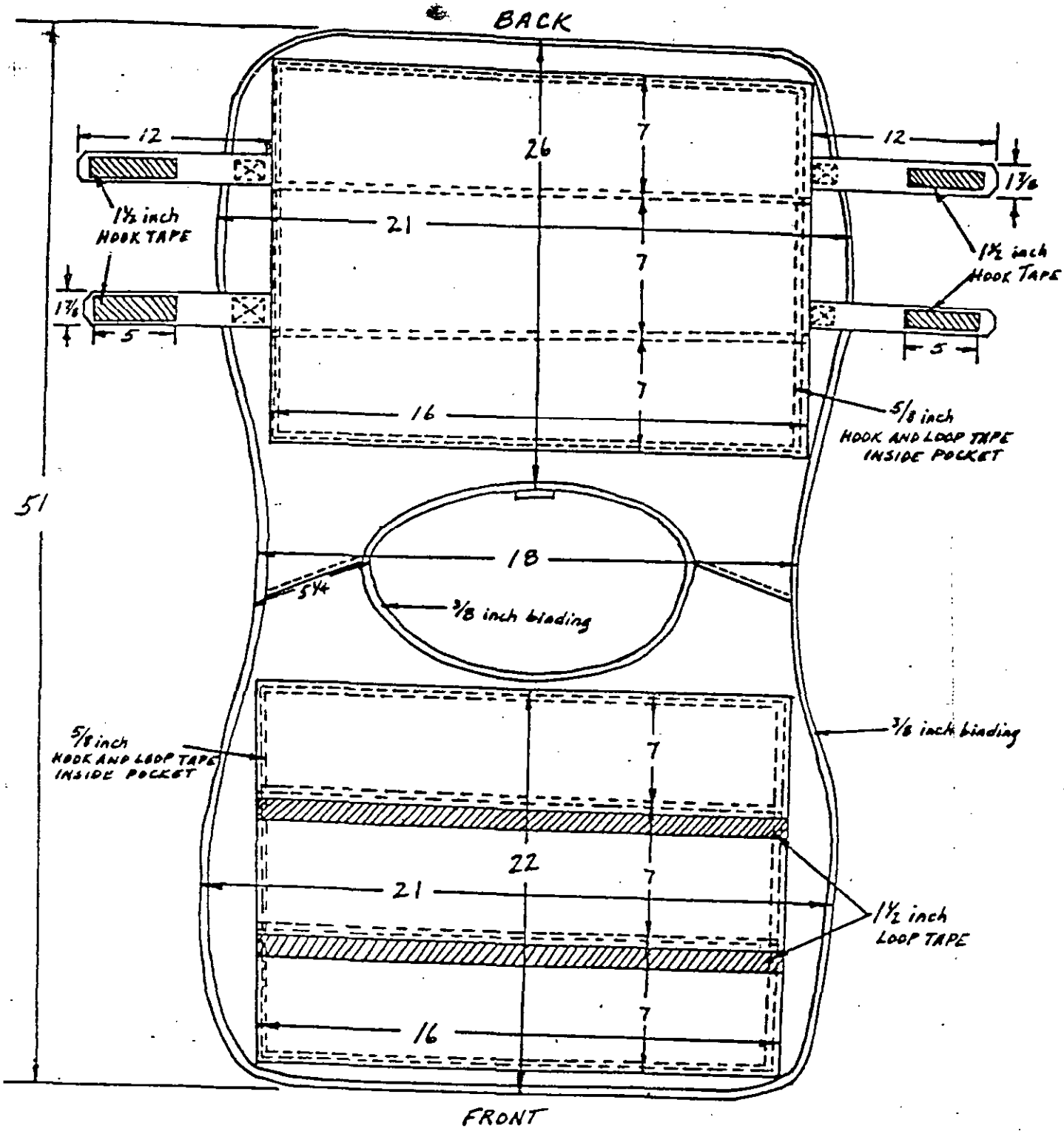
User Activity:
Coast Guard - CG

Civil Agency Coordinating Activity:
GSA - FSS

Preparing Activity:
DLA - CT

Project No. 8415-0896

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All measurements are in inches
Tolerance for all measurements is ± 5%
Hook tape should be on underside of straps as shown.

FIGURE 1 - VEST, COOLING (WITH FREEZABLE GEL STRIPS)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-50373A	2. DOCUMENT DATE (YYMMDD) 92-12-07

3. DOCUMENT TITLE
Vest, Cooling (With Freezable Gel Strips)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME Defense Personnel Support Center ATTN: DPSC-FSSD (12-3-D)	b. TELEPHONE *(Include Area Code)* (1) Commercial: 215-737-8105 (2) AUTOVON: 444-8105
c. ADDRESS *(Include Zip Code)* 2800 South 20th Street P. O. Box 8419 Philadelphia, Pennsylvania 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: *Defense Quality and Standardization Office* 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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